

Лабораторная работа №15

Настройка сетевого журналирования

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Цель работы

Получение навыков по работе с журналами системных событий и настройка централизованного сбора логов на базе службы **rsyslog**.

Задачи работы

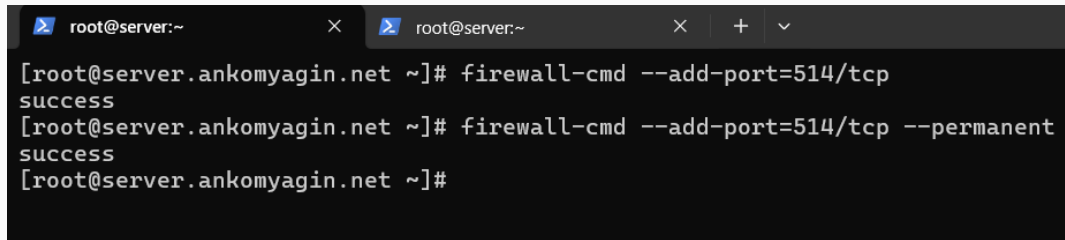
- Настройка сервера для приема логов по TCP
- Настройка клиента для отправки логов на сервер
- Анализ журналов событий
- Автоматизация настройки (Provisioning)

Настройка сервера

Создание файла конфигурации для загрузки модуля TCP и прослушивания порта 514.

```
[ankomyagin@server.ankomyagin.net client]$ cd /etc/rsyslog.d
[ankomyagin@server.ankomyagin.net rsyslog.d]$ touch netlog-server.conf
touch: cannot touch 'netlog-server.conf': Permission denied
[ankomyagin@server.ankomyagin.net rsyslog.d]$ sudo -i
[sudo] password for ankomyagin:
[root@server.ankomyagin.net ~]# touch netlog-server.conf
[root@server.ankomyagin.net ~]# nano /etc/rsyslog.d/netlog-server.conf
[root@server.ankomyagin.net ~]# |
```

Открытие порта 514/tcp для входящих соединений.

A terminal window with a dark background and light gray text. The window has two tabs at the top, both labeled 'root@server:~'. The terminal shows three lines of commands and their outputs. The first command is 'firewall-cmd --add-port=514/tcp' which returns 'success'. The second command is 'firewall-cmd --add-port=514/tcp --permanent' which also returns 'success'. The third line shows the prompt '[root@server.ankomyagin.net ~]#' without any further input or output.

```
[root@server.ankomyagin.net ~]# firewall-cmd --add-port=514/tcp
success
[root@server.ankomyagin.net ~]# firewall-cmd --add-port=514/tcp --permanent
success
[root@server.ankomyagin.net ~]#
```


Проверка того, что служба запущена и порт прослушивается.

```
[root@server.ankomyagin.net ~]# systemctl status rsyslog
● rsyslog.service - System Logging Service
   Loaded: loaded (/usr/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-15 16:35:38 UTC; 18min ago
     Invocation: 50e9d1f08f9841cab8e0blad99dc9ecf
       Docs: man:rsyslogd(8)
             https://www.rsyslog.com/doc/
    Main PID: 16651 (rsyslogd)
      Tasks: 8 (limit: 16823)
     Memory: 2.7M (peak: 3.4M)
        CPU: 1.325s
    CGroup: /system.slice/rsyslog.service
            └─16651 /usr/sbin/rsyslogd -n

Nov 15 16:35:38 server.ankomyagin.net systemd[1]: Starting rsyslog.service - System Logging Service...
Nov 15 16:35:38 server.ankomyagin.net systemd[1]: Started rsyslog.service - System Logging Service.
Nov 15 16:35:38 server.ankomyagin.net rsyslogd[16651]: [origin software="rsyslogd" swVersion="8.2412.0-1.el10" x-pid="16651" x-info="https://www.rsyslog.co
Nov 15 16:35:38 server.ankomyagin.net rsyslogd[16651]: imjournal: journal files changed, reloading... [v8.2412.0-1.el10 try https://www.rsyslog.com/e/0 ]

[root@server.ankomyagin.net ~]# netstat -tlnp | grep 514
tcp        0      0 0.0.0.0:514          0.0.0.0:*           LISTEN     16651/rsyslogd
tcp6       0      0 :::*                :::*                LISTEN     16651/rsyslogd
[root@server.ankomyagin.net ~]# ss -tlnp | grep 514
LISTEN 0      25        0.0.0.0:514          0.0.0.0:*           users:(("rsyslogd",pid=16651,fd=4))
LISTEN 0      25        [::]:514           [::]:*              users:(("rsyslogd",pid=16651,fd=5))
```

Настройка клиента

Настройка отправки всех системных событий (*.*) на удаленный сервер.

```
[root@client.ankomyagin.net server]# cd /etc/rsyslog.d
[root@client.ankomyagin.net rsyslog.d]# touch netlog-client.conf
[root@client.ankomyagin.net rsyslog.d]# nano /etc/rsyslog.d/netlog-client.conf
[root@client.ankomyagin.net rsyslog.d]# systemctl restart rsyslog
[root@client.ankomyagin.net rsyslog.d]#
```

Проверка доступности порта и отправка тестового сообщения через утилиту `logger`.

```
[root@client.ankomyagin.net rsyslog.d]# logger -p local0.info "Network logging test completed successfully"
[root@client.ankomyagin.net rsyslog.d]# telnet server.ankomyagin.net 514
Trying 192.168.1.1...
Connected to server.ankomyagin.net.
Escape character is '^['.
```

Анализ журналов

Просмотр логов на сервере

Мониторинг файла `/var/log/messages` в реальном времени. Видны сообщения от клиента.

```
[root@server.ankomyagin.net ~]# tail -f /var/log/messages
Nov 15 16:36:47 client kernel: traps: VBoxClient[35462] trap int3 ip:41dc5b sp:7fc146fb4cd0 error:0 in VBoxClient[1dc5b,400000+bb000]
Nov 15 16:36:47 client systemd-coredump[35463]: Process 35459 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov 15 16:36:47 client systemd[1]: Started systemd-coredump@2209-35463-0.service - Process Core Dump (PID 35463/UID 0).
Nov 15 16:36:48 client systemd-coredump[35464]: Process 35459 (VBoxClient) of user 1001 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10
.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm
libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 35462:#012#0 0x00000000041dc5b
n/a (n/a + 0x0)#012#1 0x00000000041dbd4 n/a (n/a + 0x0)#012#2 0x000000000450b9c n/a (n/a + 0x0)#012#3 0x0000000004359a0 n/a (n/a + 0x0)#012#4 0x0000
7fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 35460:#012#0 0x00007
fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x000000000435000 n/a (n/a + 0x0)#012#2 0x00000000045137b n/a (n/a + 0x0)#012#3 0x000000000435a3a n/a
(n/a + 0x0)#012#4 0x000000000450b9c n/a (n/a + 0x0)#012#5 0x0000000004359a0 n/a (n/a + 0x0)#012#6 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68
)#012#7 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 35461:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012
#1 0x0000000004348b2 n/a (n/a + 0x0)#012#2 0x0000000004507e6 n/a (n/a + 0x0)#012#3 0x000000000416559 n/a (n/a + 0x0)#012#4 0x0000000004182da n/a (n/a
+ 0x0)#012#5 0x000000000417d6a n/a (n/a + 0x0)#012#6 0x000000000404860 n/a (n/a + 0x0)#012#7 0x000000000450b9c n/a (n/a + 0x0)#012#8 0x00000000043
59a0 n/a (n/a + 0x0)#012#9 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#10 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack tra
ce of thread 35459:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x0000000004348b2 n/a (n/a + 0x0)#012#2 0x0000000004507e6 n/a (n/a +
0x0)#012#3 0x000000000405123 n/a (n/a + 0x0)#012#4 0x00007fc15560b30e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007fc15560b3c9 __libc_start
_main@@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x0000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Nov 15 16:36:48 client systemd[1]: systemd-coredump@2209-35463-0.service: Deactivated successfully.
Nov 15 16:36:53 client kernel: traps: VBoxClient[35473] trap int3 ip:41dc5b sp:7fc146fb4cd0 error:0 in VBoxClient[1dc5b,400000+bb000]
Nov 15 16:36:53 client systemd-coredump[35474]: Process 35470 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov 15 16:36:53 client systemd[1]: Started systemd-coredump@2210-35474-0.service - Process Core Dump (PID 35474/UID 0).
Nov 15 16:36:54 client systemd-coredump[35475]: Process 35470 (VBoxClient) of user 1001 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10
.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm
libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 35472:#012#0 0x00000000041dc5b
n/a (n/a + 0x0)#012#1 0x00000000041dbd4 n/a (n/a + 0x0)#012#2 0x000000000450b9c n/a (n/a + 0x0)#012#3 0x0000000004359a0 n/a (n/a + 0x0)#012#4 0x0000
7fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 35471:#012#0 0x00007
fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x000000000435000 n/a (n/a + 0x0)#012#2 0x00000000045137b n/a (n/a + 0x0)#012#3 0x000000000435a3a n/a
(n/a + 0x0)#012#4 0x000000000450b9c n/a (n/a + 0x0)#012#5 0x0000000004359a0 n/a (n/a + 0x0)#012#6 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68
)#012#7 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 35470:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x0000000004348b2 n/a (n/a + 0x0)#012#2 0x0000000004507e6 n/a (n/a +
0x0)#012#3 0x000000000405123 n/a (n/a + 0x0)#012#4 0x00007fc15560b30e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007fc15560b3c9 __libc_start
_main@@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x0000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Nov 15 16:36:54 client systemd[1]: systemd-coredump@2210-35474-0.service: Deactivated successfully.
Nov 15 16:36:59 client kernel: traps: VBoxClient[35484] trap int3 ip:41dc5b sp:7fc146fb4cd0 error:0 in VBoxClient[1dc5b,400000+bb000]
```

Разделение потоков сообщений от сервера и клиента с помощью grep.

```
[ankomyagin@server.ankomyagin.net ~]$ sudo -i
[sudo] password for ankomyagin:
[root@server.ankomyagin.net ~]# tail -f /var/log/messages | grep -E "(client|server)"
Nov 15 16:51:50 client systemd[1]: Started systemd-coredump[37250-0.service - Process Core Dump (PID 37250/UID 0)].
Nov 15 16:51:50 client systemd-coredump[37251]: Process 37246 (VBoxClient) of user 1001 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10
.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm
libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 37249:#012#0 0x000000000041dc5b
n/a (n/a + 0x0)#012#1 0x000000000041dbd4 n/a (n/a + 0x0)#012#2 0x0000000000450b9c n/a (n/a + 0x0)#012#3 0x00000000004359a0 n/a (n/a + 0x0)#012#4 0x0000
7fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 37248:#012#0 0x0000
7fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2 0x00000000004507e6 n/a (n/a + 0x0)#012#3 0x0000000000416559 n/a
(n/a + 0x0)#012#4 0x00000000004182da n/a (n/a + 0x0)#012#5 0x0000000000417d6a n/a (n/a + 0x0)#012#6 0x0000000000404860 n/a (n/a + 0x0)#012#7 0x00000000
00450b9c n/a (n/a + 0x0)#012#8 0x00000000004359a0 n/a (n/a + 0x0)#012#9 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#10 0x00007fc1556e66bc __
clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 37247:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2
0x000000000045137b n/a (n/a + 0x0)#012#3 0x0000000000435a3a n/a (n/a + 0x0)#012#4 0x0000000000450b9c n/a (n/a + 0x0)#012#5 0x000000000043
59a0 n/a (n/a + 0x0)#012#6 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#7 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack tra
ce of thread 37246:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2 0x00000000004507e6 n/a (n/a +
0x0)#012#3 0x0000000000405123 n/a (n/a + 0x0)#012#4 0x00007fc1556e630e __libc_start_call_main (libc.so.6 + 0x2a30e)#012#5 0x00007fc1556e63c9 __libc_start
_main@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary architecture: AMD x86-64
Nov 15 16:51:50 client systemd[1]: systemd-coredump[37267-0.service: Deactivated successfully.
Nov 15 16:51:50 server kernel: traps: VBoxClient[22656] trap int3 ip:41dc5b sp:7fc146fb4cd0 error:0 in VBoxClient[1dc5b,400000+bb000]
Nov 15 16:51:50 server systemd-coredump[22657]: Process 22653 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov 15 16:51:50 server systemd[1]: Started systemd-coredump[161-22657-0.service - Process Core Dump (PID 22657/UID 0)].
Nov 15 16:51:51 server systemd-coredump[22658]: Process 22653 (VBoxClient) of user 1001 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10
.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm
libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 22656:#012#0 0x000000000041dc5b
n/a (n/a + 0x0)#012#1 0x000000000041dbd4 n/a (n/a + 0x0)#012#2 0x0000000000450b9c n/a (n/a + 0x0)#012#3 0x00000000004359a0 n/a (n/a + 0x0)#012#4 0x0000
7fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 22655:#012#0 0x0000
7fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2 0x00000000004507e6 n/a (n/a + 0x0)#012#3 0x0000000000416559 n/a
(n/a + 0x0)#012#4 0x00000000004182da n/a (n/a + 0x0)#012#5 0x0000000000417d6a n/a (n/a + 0x0)#012#6 0x0000000000404860 n/a (n/a + 0x0)#012#7 0x00000000
00450b9c n/a (n/a + 0x0)#012#8 0x00000000004359a0 n/a (n/a + 0x0)#012#9 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#10 0x00007fc1556e66bc __
clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 22653:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2
0x00000000004507e6 n/a (n/a + 0x0)#012#3 0x0000000000405123 n/a (n/a + 0x0)#012#4 0x00007fc1556e630e __libc_start_call_main (libc.so.6 + 0
x2a30e)#012#5 0x00007fc1556e63c9 __libc_start_main@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary archite
cture: AMD x86-64
Nov 15 16:51:51 server systemd[1]: systemd-coredump[161-22657-0.service: Deactivated successfully.
Nov 15 16:51:51 server systemd[1]: Starting systemd-hostnamed.service - Hostname Service...
Nov 15 16:51:51 server systemd[1]: Started systemd-hostnamed.service - Hostname Service.
Nov 15 16:51:55 client kernel: traps: VBoxClient[37260] trap int3 ip:41dc5b sp:7fc146fb4cd0 error:0 in VBoxClient[1dc5b,400000+bb000]
Nov 15 16:51:55 client systemd-coredump[37261]: Process 37257 (VBoxClient) of user 1001 terminated abnormally with signal 5/TRAP, processing...
Nov 15 16:51:55 client systemd[1]: Started systemd-coredump[37261-0.service - Process Core Dump (PID 37261/UID 0)].
Nov 15 16:51:55 client systemd-coredump[37262]: Process 37257 (VBoxClient) of user 1001 dumped core.#012#012Module libXau.so.6 from rpm libXau-1.0.11-8.el10
.x86_64#012Module libxcb.so.1 from rpm libxcb-1.17.0-3.el10.x86_64#012Module libX11.so.6 from rpm libX11-1.8.10-1.el10.x86_64#012Module libffi.so.8 from rpm
libffi-3.4.4-9.el10.x86_64#012Module libwayland-client.so.0 from rpm wayland-1.23.0-2.el10.x86_64#012Stack trace of thread 37260:#012#0 0x000000000041dc5b
n/a (n/a + 0x0)#012#1 0x000000000041dbd4 n/a (n/a + 0x0)#012#2 0x0000000000450b9c n/a (n/a + 0x0)#012#3 0x00000000004359a0 n/a (n/a + 0x0)#012#4 0x0000
7fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#5 0x00007fc1556e66bc __clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 37259:#012#0 0x0000
7fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2 0x00000000004507e6 n/a (n/a + 0x0)#012#3 0x0000000000416559 n/a
(n/a + 0x0)#012#4 0x00000000004182da n/a (n/a + 0x0)#012#5 0x0000000000417d6a n/a (n/a + 0x0)#012#6 0x0000000000404860 n/a (n/a + 0x0)#012#7 0x00000000
00450b9c n/a (n/a + 0x0)#012#8 0x00000000004359a0 n/a (n/a + 0x0)#012#9 0x00007fc155675b68 start_thread (libc.so.6 + 0x94b68)#012#10 0x00007fc1556e66bc __
clone3 (libc.so.6 + 0x1056bc)#012#012Stack trace of thread 37257:#012#0 0x00007fc1556e44bd syscall (libc.so.6 + 0x1034bd)#012#1 0x00000000004348b2 n/a (n/a + 0x0)#012#2
0x00000000004507e6 n/a (n/a + 0x0)#012#3 0x0000000000405123 n/a (n/a + 0x0)#012#4 0x00007fc1556e630e __libc_start_call_main (libc.so.6 + 0
x2a30e)#012#5 0x00007fc1556e63c9 __libc_start_main@GLIBC_2.34 (libc.so.6 + 0x2a3c9)#012#6 0x00000000004044aa n/a (n/a + 0x0)#012ELF object binary archite
cture: AMD x86-64
```

Автоматизация

Подготовка конфигурационных файлов и создание скриптов для автоматического развертывания через Vagrant.

```
[root@server.ankomyagin.net ~]# cd /vagrant/provision/server
mkdir -p /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.ankomyagin.net server]# cp -R /etc/rsyslog.d/netlog-server.conf /vagrant/provision/server/netlog/etc/rsyslog.d
[root@server.ankomyagin.net server]# cd /vagrant/provision/server
[root@server.ankomyagin.net server]# touch netlog.sh
[root@server.ankomyagin.net server]# chmod +x netlog.sh
[root@server.ankomyagin.net server]# nano netlog.sh
[root@server.ankomyagin.net server]# |
```

Контрольные вопросы

Модуль для journald

- imjournal

Устаревший модуль для локального сокета

- imuxsock

Отключение устаревшего метода

- Параметр `$OmitLocalLogging on`

Основной файл конфигурации

- `/etc/rsyslog.conf`

Включение приема по TCP Необходимо добавить директивы:

```
$ModLoad imtcp
```

```
$InputTCPServerRun 514
```

Настройка Firewall Команда для разрешения приема логов:

```
firewall-cmd --add-port=514/tcp --permanent
```

Выводы

- Настроен сервер централизованного сбора логов.
- Организована передача событий с клиента по протоколу TCP.
- Отработаны навыки фильтрации и анализа системных журналов.
- Созданы скрипты для автоматизации процесса.